

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION SP/2024)

CLASS: BTECH/IMSC.
BRANCH: BT/CHEMICAL/CIVIL/MECH/PIE/FT/PHYSICS

SEMESTER : II
SESSION : SP/2024

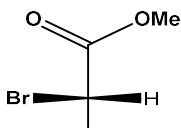
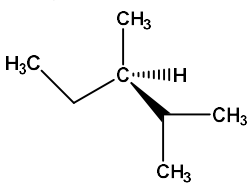
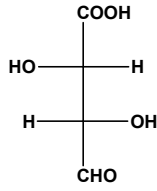
SUBJECT: CH101 CHEMISTRY

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

		CO	BL
Q.1(a)	Calculate crystal field stabilization energy (CFSE) for the following complexes: (a) $[\text{Fe}(\text{CN})_6]^{3-}$ (b) $[\text{CoCl}_6]^{4-}$	[2] 1	Apply
Q.1(b)	Explain bonding in $[\text{CoF}_6]^{3-}$ complex using Valence bond theory.	[3] 1	Understand
Q.2(a)	Explain charge transfer phenomenon in complexes with example.	[2] 1	Understand
Q.2(b)	Discuss crystal field splitting of d orbitals in tetrahedral complexes with diagram. Why splitting of tetrahedral complex is less in comparison to octahedral complex ?	[3] 1	Understand
Q.3(a)	Discuss how the ease of formation of cyclic compounds differs for <i>cis</i> and <i>trans</i> isomers with example.	[2] 2	Understand
Q.3(b)	Give the schematic representation of polarimeter .	[3] 2	Remember
Q.4(a)	Compare between conformation and configuration with example.	[2] 2	Evaluate
Q.4(b)	Assign R and S to the following:	[3] 2	Evaluate
	 (i)  (ii)  (iii)		
Q.5(a)	Explain how substituted allenes can be optically active, inspite of having no chiral carbon.	[2] 2	Understand
Q.5(b)	Explain z-in and z-out Jahn Teller distortion.	[3] 1	Understand

:21/02/2024:E